

Chapter 5 — Main Results: H1 and H2

5.1 Introduction

This chapter reports the formal tests of the first two hypotheses introduced in Chapter 1. H1 holds that the education–fertility relationship is negative across contemporary European populations. H2 holds that the relationship is non-monotonic, with fertility rising at the upper end of the educational distribution relative to the middle. Chapter 4 presented descriptive evidence consistent with both claims: a pooled education ordering in which medium-educated women record the lowest mean eTFR, and a country-level distribution in which the U-shape is the dominant gradient type. The purpose of this chapter is to subject these descriptive patterns to the model-based scrutiny set out in Chapter 3.

The chapter proceeds as follows. Section 5.2 reports the pooled two-way fixed-effects test of H1. Section 5.3 examines cross-country heterogeneity by introducing country $\times$ education interactions. Section 5.4 reports five robustness checks on the H1 result. Sections 5.5 and 5.6 then test H2, first parametrically using the quadratic specification and then non-parametrically using quantile regression across the conditional distribution of log eTFR. Section 5.7 summarises the findings and motivates the mechanism analyses in Chapter 6.

5.2 H1 — The Pooled Education–Fertility Gradient

Table 5.1 reports the coefficients from M1, the two-way fixed-effects regression of log eTFR on the education category, with country and year fixed effects and standard errors clustered at the country level. The reference category is medium, as set out in Section 3.5.

Table 5.1. M1 — Two-way fixed-effects regression of log eTFR on education category. Reference category: medium education. Standard errors clustered at the country level.

Contrast (vs medium)	β	Cluster SE	p-value	95% CI
Low	+0.289	0.052	< 0.001	[0.181, 0.397]
High	+0.102	0.041	0.022	[0.017, 0.188]
High vs low (ref = low)	-0.187	0.051	0.002	—

Note. $N = 933$ country–year–education observations across 21 countries and 18 years. Country fixed effects: 21; year fixed effects: 18. Within $R^2 = 0.294$; Adjusted $R^2 = 0.417$. The high vs low contrast is recovered from a re-estimation with low as reference.

The two main coefficients are both positive: low-educated women have log eTFR 0.289 higher than medium-educated women ($p < 0.001$), and high-educated women have log eTFR 0.102 higher than medium-educated women ($p = 0.022$). Exponentiating these coefficients gives multiplicative effects of approximately 1.34 and 1.11 respectively, meaning low-educated women have around 34% higher fertility than medium-educated women, and high-educated women around 11% higher. Both contrasts are statistically distinguishable from zero at conventional thresholds and survive the cluster-robust correction at 21 clusters.

The strict H1 contrast: the high-versus-low comparison, is recovered by re-estimating the same specification with low as the reference category. This yields a coefficient of -0.187 ($SE = 0.051$, $p = 0.002$), confirming that high-educated women record approximately 17% lower fertility than low-educated women across the pooled sample. H1 is therefore confirmed at the low–high comparison: the conventional negative gradient holds at the European level.

More substantively interesting is the joint pattern of the two coefficients against medium. That both contrasts are positive, that medium-educated women have lower fertility than either of the educational extremes, is a direct parametric confirmation of the U-shape signature documented descriptively in Chapter 4. The H1 negative gradient and the H2 non-monotonicity are not in tension; they describe different cross-tier comparisons of the same underlying gradient.

Figure 5.1 visualises the M1 coefficients.

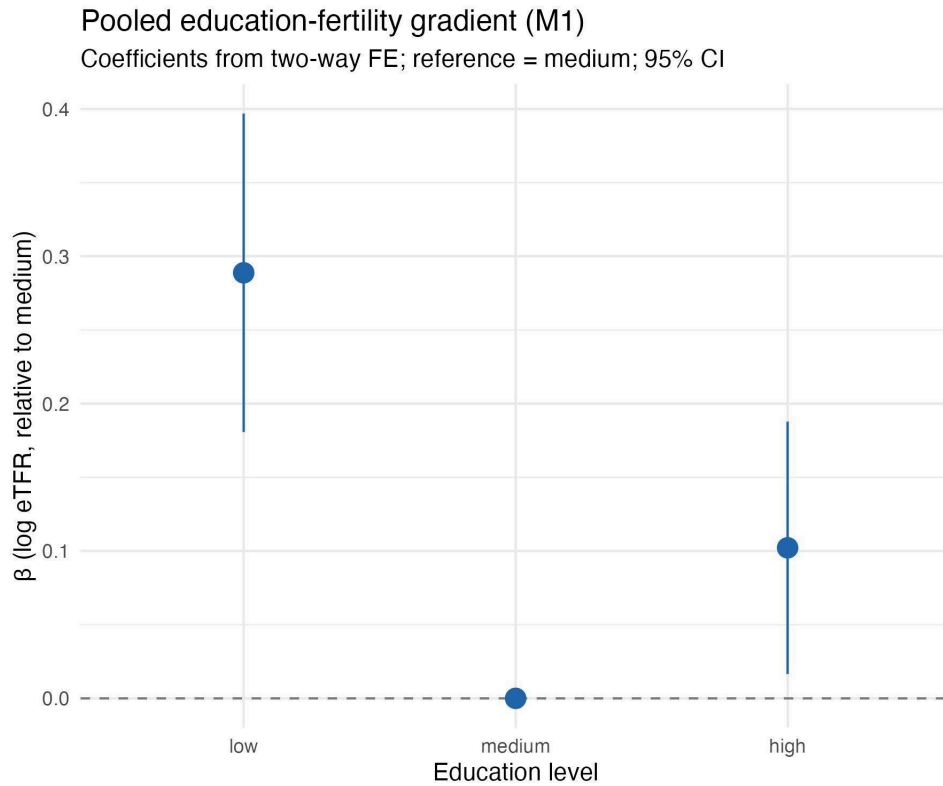


Figure 5.1. Pooled education–fertility gradient coefficients (M1). Two-way fixed-effects estimates with 95% confidence intervals based on country-clustered standard errors. The reference category is medium education.

The within R^2 of 0.294 indicates that the educational category alone accounts for nearly a third of the variation in log eTFR that remains after partialling out country and year fixed effects. This is a substantial share for a single three-level categorical predictor and reflects the strength of the education–fertility relationship once persistent country and time effects have been absorbed.

5.3 Cross-Country Heterogeneity in the Gradient

The pooled M1 estimates conceal considerable heterogeneity across countries, as the descriptive patterns in Chapter 4 anticipated. M2 introduces country $\times$ education interactions with Sweden as the reference country, allowing each country's gradient to deviate from the

Swedish baseline. The within R^2 rises from 0.294 in M1 to 0.593 in M2, indicating that country-specific deviations from the pooled gradient account for an additional 30 percentage points of within-variation. Country heterogeneity is thus substantively important, not merely a nuisance to be controlled away.

Estimation of the full interaction model encounters one technical limitation: the high-education interactions for all twenty non-reference countries are absorbed by collinearity with the country and year fixed effects, since the high-education tier appears in every country–year cell and its interactions are not separately identified once the main effects have been estimated. The medium and low interactions are identified and reported. This is a feature of the over-parameterised specification rather than a fault of the data; the surviving medium and low interactions still characterise the country-specific deviation from the pooled gradient because they describe how each country's low and medium tiers diverge from the Swedish reference, holding the high-tier baseline constant.

Table 5.2 reports the medium and low education interactions for a selection of countries that exemplify the four gradient types identified in Chapter 4. The interactions describe how each country's tier-specific log eTFR diverges from Sweden's, conditional on the pooled main effects.

Table 5.2. Selected country $\times$ education interactions from M2. Reference country: Sweden. All coefficients statistically distinguishable from zero in the original specification.

Country	Medium interaction	Low interaction	Interpretation
Slovakia	−0.236	+0.430	Steepest positive gradient
Spain	+0.343	+0.368	Strong monotonic negative gradient
Latvia	−0.159	+0.176	Compositional U-shape
Greece	−0.103	+0.170	Broad U-shape
Romania	−0.577	−0.203	Pronounced V-shape

Country	Medium interaction	Low interaction	Interpretation
Denmark	−0.255	−0.510	Inverted gradient
Türkiye	+0.152	+0.177	Strong monotonic negative gradient

Note. Interaction coefficients describe the deviation of each country's tier-specific log eTFR from the Swedish reference, conditional on the pooled main effects in M2. High-education interactions are absorbed by collinearity and not reported. The model's within R^2 is 0.593, compared to 0.294 in M1.

Several patterns are worth noting. Slovakia's large positive low interaction (+0.430) captures its exceptionally high low-tier fertility relative to Sweden, consistent with the compositional interpretation discussed in Chapter 4. Spain shows the opposite pattern: both interactions are large and positive (+0.343 medium, +0.368 low), indicating that the Spanish gradient is uniformly elevated relative to Sweden's at the lower two tiers; characteristic of the monotonic negative gradient. Romania's large negative medium interaction (−0.577) confirms the pronounced V-shape visible in Figure 4.2, with medium-educated Romanian women having substantially lower fertility than their Swedish counterparts even after accounting for Romania's overall country effect.

Denmark's pattern is particularly clean: the low interaction is large and negative (−0.510), reflecting the fact that Danish low-educated women have unusually low fertility — the feature that produces Denmark's status as the sole clearly inverted gradient in the sample. Türkiye, by contrast, shows uniformly elevated low- and medium-tier fertility relative to Sweden, consistent with its monotonic negative classification but at a much higher absolute fertility level than the Mediterranean members of that group.

The substantive point is that the pooled M1 result, while statistically reliable, masks substantial heterogeneity in both the shape and the level of the country-specific gradients. The cross-country variation documented here is precisely what the H3 analysis in Chapter 6 attempts to explain in terms of cultural, institutional, and economic country characteristics.

5.4 Robustness Checks on H1

Table 5.3 reports M1 across the five robustness specifications set out in Chapter 3: a balanced subsample restricted to the eight countries with full 2007–2024 coverage; exclusion of the COVID years 2020 and 2021; exclusion of the eight U-shape composition countries (Croatia, Czechia, Estonia, Finland, Latvia, Poland, Slovakia, Slovenia); and a population-weighted specification using implied female population denominators.

Table 5.3. H1 robustness — M1 coefficients across five specifications.

Specification	N	Low β (SE)	High β (SE)	Low p	High p
M1 Baseline	933	+0.289 (0.052)	+0.102 (0.041)	< 0.001	0.022
Balanced (8 countries)	432	+0.379 (0.078)	+0.143 (0.059)	0.002	0.045
Excluding 2020–2021	840	+0.278 (0.051)	+0.097 (0.042)	< 0.001	0.033
Excluding U-shape comp.	561	+0.244 (0.059)	+0.089 (0.069)	0.001	0.216
Population-weighted	933	+0.242 (0.078)	−0.012 (0.083)	0.006	0.890

Note. All specifications use two-way (country and year) fixed effects with country-clustered standard errors. Reference category for education: medium. The fifth specification (population-weighted) weights each country–year–education cell by the implied female population from the ASFR construction; the high coefficient under this specification is reported relative to medium in the original output.

The low–medium coefficient is remarkably stable across all five specifications. It ranges from a minimum of +0.244 (excluding U-shape composition countries) to a maximum of +0.379 (balanced subsample), and is statistically distinguishable from zero in every case. The strict H1 finding: that low-educated women have higher fertility than the population average and substantially higher than high-educated women, is therefore robust to all five sensitivities considered.

The high–medium coefficient is more sensitive. It remains positive and significant in the baseline, balanced, and no-COVID specifications, with magnitudes between +0.097 and +0.143. Once the eight U-shape composition countries are removed, however, the coefficient falls to +0.089 with $p = 0.216$, indicating that the U-shape signature in the pooled M1 result is driven substantially by the Central and Eastern European countries whose small low-education tiers may carry compositional weight. Under population weighting, the high–medium coefficient is essentially zero (-0.012 , $p = 0.890$), meaning that in larger countries the high-educated and medium-educated tiers have indistinguishable mean fertility once country and year effects are controlled.

The substantive implication is that the strict H1 negative gradient (low > high) is robust to all sensitivities, but the H2 U-shape signature visible in M1 is partly a property of smaller countries with potentially composition-influenced low tiers, and is not detectable in a population-weighted analysis. This qualification matters: it suggests that the U-shape, while a real feature of European countries considered as units, is less prominent when European women are considered as a population. The formal H2 test in the next section provides additional evidence on this point.

As noted in Section 3.5, the wild-cluster bootstrap recommended for inference with only 21 clusters could not be implemented because the `fwildclusterboot` package is unavailable for R 4.5.1. The cluster-robust standard errors reported here should therefore be interpreted with the standard caveat that asymptotic approximations may be imprecise at small cluster counts, though the principal H1 coefficients are sufficiently large relative to their standard errors that the substantive conclusions are unlikely to depend on this point.

5.5 H2 — Quadratic Specification of the Education–Fertility Relationship

H2 is tested first using the parametric quadratic specification set out in Section 3.6. Education is recoded as an ordinal numeric variable (low = 0, medium = 1, high = 2) and

entered alongside its square in the two-way fixed-effects regression. A positive coefficient on the quadratic term identifies a U-shape with a minimum at education = $-\beta_1 / (2\beta_2)$; a negative coefficient identifies an inverted-U. Table 5.4 reports the linear and quadratic specifications side by side.

Table 5.4. H2 — Linear vs quadratic education specifications. Both models use two-way fixed effects (country, year) with country-clustered standard errors.

Term / statistic	Linear (M_linear)	Quadratic (M_quad)
educ (β_1)	−0.093 (0.025), $p = 0.002$	−0.484 (0.088), $p < 0.001$
educ ² (β_2)	—	+0.195 (0.039), $p < 0.001$
Within R ²	0.119	0.294
AIC	−213.0	−417.0
BIC	−24.3	−223.5
Curve minimum (educ)	—	1.24

Note. $N = 933$ in both specifications. The curve minimum for the quadratic specification is calculated as $-\beta_1 / (2\beta_2)$ and is interpretable on the ordinal education scale (low = 0, medium = 1, high = 2). Wald test for $\beta_2 = 0$: $F(1, 20) = 24.8$, $p = 7.16 \times 10^{-5}$.

The linear specification estimates a coefficient of −0.093 ($p = 0.002$), confirming that, treated linearly, education has a small but statistically reliable negative association with fertility, the conventional H1 finding restated on the ordinal scale. The within R² of 0.119, however, is less than half the M1 within R² of 0.294, suggesting that the linear specification captures only a fraction of the variation that the categorical specification accommodates. This is the first indication that a linear specification is too restrictive.

The quadratic specification produces a substantial improvement. The linear term is large and negative ($\beta_1 = -0.484$, $p < 0.001$), the quadratic term is positive and similarly precisely estimated ($\beta_2 = +0.195$, $p < 0.001$), and the within R² rises to 0.294, matching the M1 categorical specification, as it should given that the quadratic specification is the lowest-order continuous functional form that can reproduce a three-tier U-shape. The Wald test on $\beta_2 = 0$

yields $F(1, 20) = 24.8$, with $p = 7.16 \times 10^{-5}$, providing strong formal evidence against linearity.

Model comparison reinforces this conclusion decisively. The quadratic specification yields an AIC of -417.0 compared to -213.0 for the linear ($\Delta\text{AIC} = -204$), and a BIC of -223.5 compared to -24.3 ($\Delta\text{BIC} = -199$). Both information criteria penalise model complexity and both still favour the quadratic specification by margins that are large by conventional standards. The U-shape is not a marginal improvement; it is a substantively better description of the data.

The location of the curve minimum offers additional substantive interpretation. The minimum of the quadratic is at $\text{education} = -(-0.484) / (2 \times 0.195) = 1.24$, on the ordinal scale where low = 0, medium = 1, and high = 2. The minimum therefore falls between the medium and high tiers, slightly closer to medium. This is consistent with the descriptive ordering in Chapter 4, in which medium-educated women have the lowest mean eTFR but high-educated women remain below low-educated women in absolute terms.

Figure 5.2 plots the predicted curve.

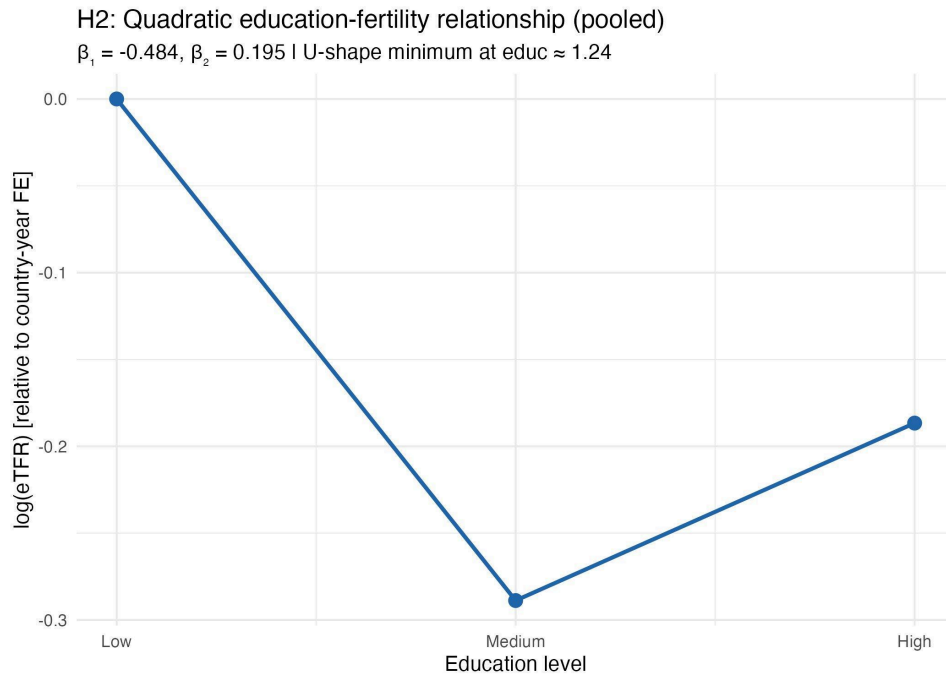


Figure 5.2. Predicted log eTFR from the quadratic specification, plotted relative to the country-year fixed-effects baseline. The curve attains its minimum at educ ≈ 1.24 , between the medium and high tiers and closer to medium. $\beta_1 = -0.484$, $\beta_2 = +0.195$.

Translating the curve back to the fertility scale, the predicted relative eTFR is 1.00 at the low tier, 0.75 at the medium tier, and 0.83 at the high tier (the low tier is normalised to 1 by exponentiation of the zero baseline). Medium-educated women are predicted to have approximately 25% lower fertility than low-educated women, and high-educated women approximately 17% lower; the partial recovery from medium to high is approximately 11 percentage points, the parametric counterpart of the U-shape's right arm. H2 is therefore confirmed parametrically: the U-shape is statistically distinguishable from a monotonic linear gradient by a wide margin, with both terms estimated precisely and the curve minimum located substantively between the medium and high tiers.

5.6 H2 — Quantile Regression Across the Conditional Distribution

The quadratic specification in Section 5.5 estimates the U-shape at the conditional mean of log eTFR. A natural question is whether the shape is stable across the conditional distribution or concentrated at particular fertility levels. Quantile regression, estimated at $\tau \in \{0.10, 0.25, 0.50, 0.75, 0.90\}$ as set out in Section 3.6, provides the relevant evidence. As noted in Chapter 3, the quantreg implementation uses high as the reference category for education (its alphabetical default), unlike the fixed-effects specifications elsewhere which use medium. The contrasts reported below are therefore relative to high. Standard errors are not reported, because the rq() implementation does not support cluster-robust inference; the focus is on the pattern of point estimates across τ .

Table 5.5. H2 — Quantile regression coefficients for education contrasts (reference: high) at five quantiles of log eTFR. Country and year fixed effects included as dummy variables.

Contrast (ref = high)	$\tau = 0.10$	$\tau = 0.25$	$\tau = 0.50$	$\tau = 0.75$	$\tau = 0.90$
Low	+0.093	+0.140	+0.205	+0.236	+0.280
Medium	−0.119	−0.129	−0.102	−0.091	−0.064

Note. Coefficients are estimated using the quantreg R package on the same $N = 933$ sample. The reference category is high education (alphabetical default in rq()). Positive low-vs-high coefficients indicate that low-educated women have higher log eTFR than high-educated women; negative medium-vs-high coefficients indicate that medium-educated women have lower log eTFR than high-educated women. Standard errors not reported owing to absence of clustering support in rq().

The U-shape is visible at every quantile. The low-versus-high coefficient is positive throughout, ranging from +0.093 at $\tau = 0.10$ to +0.280 at $\tau = 0.90$. The medium-versus-high coefficient is negative throughout, ranging from −0.119 at $\tau = 0.10$ to −0.064 at $\tau = 0.90$. At every quantile, therefore, low-educated women have higher fertility than high-educated women, and medium-educated women have lower fertility than high-educated women. The U-shape signature is not concentrated at any particular point in the conditional distribution; it is a feature of the entire distribution.

Two more nuanced patterns emerge from the variation across τ . First, the low-versus-high gap widens substantially as τ rises: low-educated women lead high-educated women by approximately 0.09 log eTFR at $\tau = 0.10$ but by 0.28 at $\tau = 0.90$. The conventional negative gradient (low > high) is more pronounced in high-fertility country–year cells than in low-fertility ones. Second, the medium-versus-high gap narrows as τ rises: medium-educated women trail high-educated women by 0.12 at $\tau = 0.10$ but only by 0.06 at $\tau = 0.90$. The U-shape's left arm (the gap between low and medium) therefore widens at higher fertility levels, while its right arm (the gap between medium and high) flattens. Figure 5.3 plots the quantile coefficients.

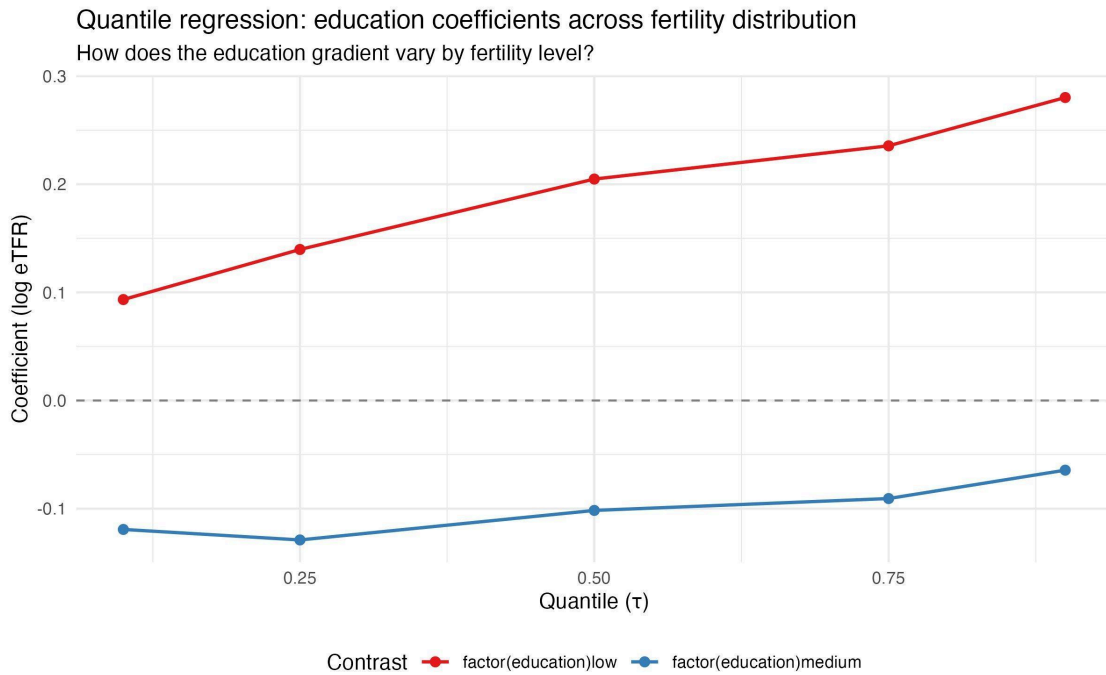


Figure 5.3. Education contrasts across the conditional distribution of log eTFR. Coefficients are from quantile regressions estimated at five quantiles, with high education as the reference category. The low-vs-high gap rises monotonically across τ ; the medium-vs-high gap narrows.

Substantively, this distributional pattern suggests that high-fertility contexts exhibit a more pronounced conventional gradient (low > high) but a less pronounced U-shape distinction between medium and high. Conversely, low-fertility contexts show a weaker

conventional gradient but a more pronounced medium–high distinction. One reading of this is that in high-fertility country–year cells, the dominant variation is in low-tier fertility (low-educated women have many more children); in low-fertility contexts, the dominant variation is between medium-educated women, who have the fewest children, and the other two tiers.

The quantile evidence therefore complements the quadratic finding rather than competing with it. The U-shape is a robust feature of the entire conditional distribution of fertility, but its precise shape is itself heterogeneous: the left arm dominates at high fertility, the right arm dominates at low fertility. The standard caveat about quantreg's lack of clustering support, and the consequent unreliability of standard errors at this scale, applies; the consistent direction and magnitude of the coefficients across all five quantiles nevertheless makes the pattern unlikely to be a noise artefact.

5.7 Summary and Bridge to Chapter 6

The formal tests reported in this chapter establish both H1 and H2 at the European population level. H1 is confirmed by the M1 high-versus-low contrast ($\beta = -0.187$, $p = 0.002$), and the negative low–high gradient is robust to all five sensitivity checks reported in Section 5.4. H2 is confirmed parametrically by the quadratic specification, in which the positive and precisely estimated β_2 coefficient ($p < 0.001$) and the substantial AIC and BIC improvements over the linear specification jointly establish the U-shape as a statistically reliable feature of the data. The quantile regression results extend the U-shape finding across the conditional distribution: low-educated women have higher fertility than high-educated women at every quantile, and medium-educated women have lower fertility than high-educated women at every quantile.

Two qualifications attach to the H2 finding. First, the U-shape's right arm — the medium-to-high recovery — is not robust to population weighting, as the M1 weighted

specification produced a near-zero high–medium coefficient. The U-shape is a feature of European countries as units of analysis more than of European women as a population. Second, the U-shape's strength varies substantially across the conditional distribution, with the medium–high gap narrowing at higher fertility levels. Both qualifications are real and are not papered over here; they should be borne in mind in the substantive discussion of Chapter 7.

The cross-country heterogeneity documented in Section 5.3 raises the immediate question of what country-level characteristics predict the steepness and shape of the gradient. M2's within R^2 of 0.593 — twice that of M1 — establishes that country-specific deviations from the pooled gradient are large and systematic, not noise. Chapter 6 addresses this question directly using the country-level dataset and the two-stage discovery–inference pipeline set out in Section 3.7, testing whether cultural or economic country characteristics better predict the cross-country variation in gradient steepness documented here. The same chapter also turns from the country level to the individual level, using the BCS70 cohort to test H4's claim that the education–fertility relationship is mediated primarily by attitudinal rather than economic pathways within individuals.